
DV2626/DV2599 Machine Learning

Lecture 3: Data Preprocessing

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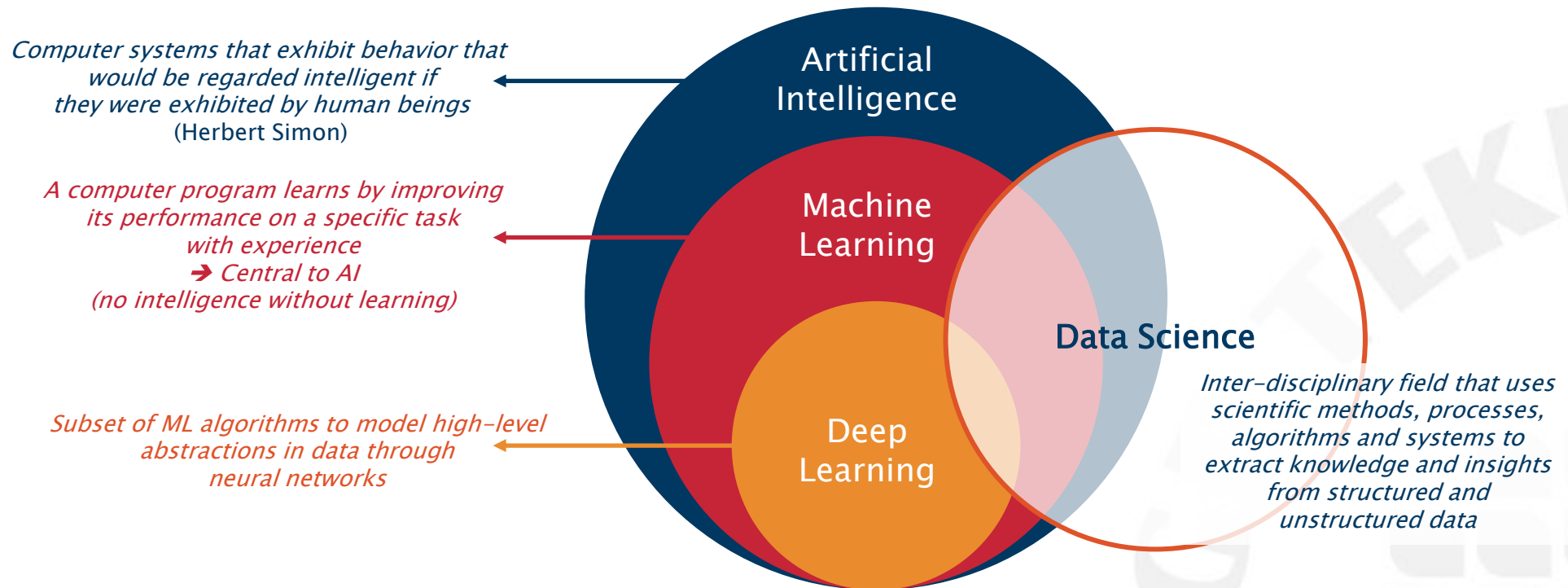
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Outline

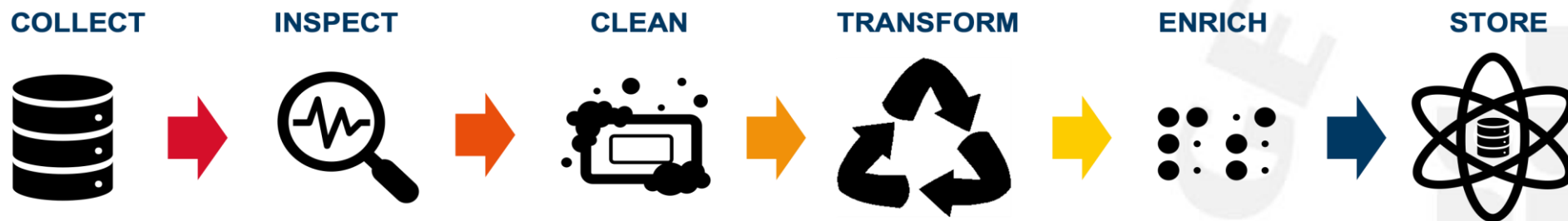
- Data Science versus AI
- Why is necessary to do data preprocessing?
- Forms of data preprocessing
 - Cleaning
 - Integration
 - Transformation
 - Reduction
 - Discretization

Data Science versus AI



Improving the code versus improving the data

- Andrew Ng (co-founder Coursera & deeplearning.ai), 06/04/2021:
 - *When a system isn't performing well, many teams instinctually try to improve the code. But for many practical applications, it's more effective instead to focus on improving the data, ...*
 - *If 80 percent of our work is data preparation, then why are we not ensuring data quality is of the utmost importance for a machine learning team ...*

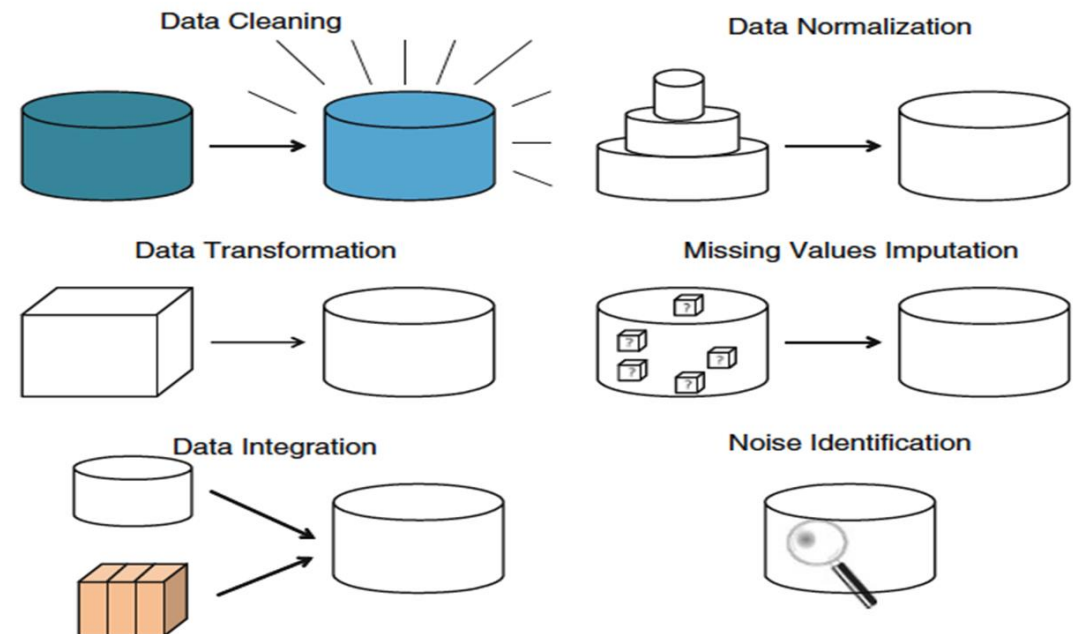


Why is necessary to do data preprocessing?

- Data in the real world is dirty
 - incomplete: lacking attribute values, lacking certain attributes of interest, or containing only aggregate data
 - noisy: containing errors or outliers
 - inconsistent: containing discrepancies in codes or names
- Most of data analysis techniques require complete data sets
- Quality decisions must be based on quality data
- Use of all available data about a studied phenomenon requires consistent data integration

Forms of data preprocessing (1)

- Data cleaning
 - Impute missing values, smooth noisy data, identify or remove outliers, and resolve inconsistencies
- Data integration
 - Integration of multiple databases or files
- Data transformation
 - Data is transformed or consolidated into forms appropriate for mining by performing summary or aggregation operations
- Data reduction
 - Obtains reduced representation in volume but produces the same or similar analytical results
- Data discretization
 - Part of data reduction but with particular importance, especially for numerical data



Data Cleaning

- Data cleaning tasks?
 - Impute missing values
 - Identify outliers and smooth out noisy data
 - Correct inconsistent data

Missing values may be due to

- Equipment malfunction
- Experimental errors
- Inconsistent with other recorded data and thus deleted
- Data not entered due to misunderstanding
- Certain data may not be considered important at the time of entry
- Not register history or changes of the data

Missing value imputation solutions

- Remove any data object (or observation) with missing values
- Repeat the experiment
- Replace the missing value by zero
- Replace the missing value by the attribute mean
- Replace the missing value by the attribute mean for all samples belonging to the same class
- Methods considering the correlation structure of the data

Missing value estimation methods

- **Local** imputation methods, e.g., KNNimpute, versus **global** methods (Bayesian approaches)
- **Single** imputation methods replace a missing value by one value, whereas in **multiple imputation**, several values are imputed
- Imputation methods based on a **single dataset** versus methods that use **multiple data sets** to estimate the missing value

KNNimpute Algorithm

Weighted K-Nearest Neighbours

feature	1			t			n
object i				y_{it}			
object m							

-- Missing value

- Consider an object (feature vector) i with a missing value at point t :

$$y_i = (y_{i1}, \dots, y_{it-1}, _, y_{it+1}, \dots, y_{in})$$

- Calculate Euclidean distance $d_e(y_i, y_j)$ for each j , which has a value at point t ;
- Rank the objects in increasing order of d_e ;
- Select the first K objects: $i_1, i_2, \dots, i_k$

- Estimate:** $y_{it} = \sum_{j=i_1}^{i_k} w_j y_{jt}$

y_{jt} : value at point t from object j

w_j : weight of object j based on its Euclidean distance.

Data are noisy

- **Noise**: random error or variance in a measured variable
- **Incorrect attribute values** may be due to
 - faulty data collection instruments
 - data entry problems
 - data transmission problems
 - technology limitation
 - inconsistency in naming convention
- **Other data problems** which require data cleaning
 - duplicate records
 - inconsistent data
 - incomplete data

Data smoothing and outlier filtering solutions

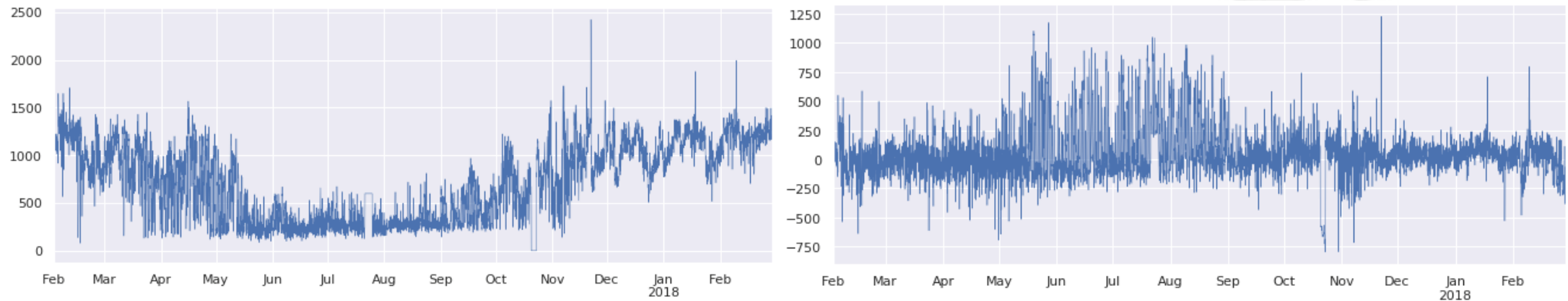
- Binning
 - first sort data and partition into (equi-depth) bins
 - then smooth data by bin mean/median/boundaries, etc.
- Regression
 - smooth by fitting the data into regression functions
- Clustering analysis
 - detect and remove outliers
- Combined computer and human inspection
 - detect suspicious values and check by human (e.g., deal with possible outliers)

Data smoothing via binning

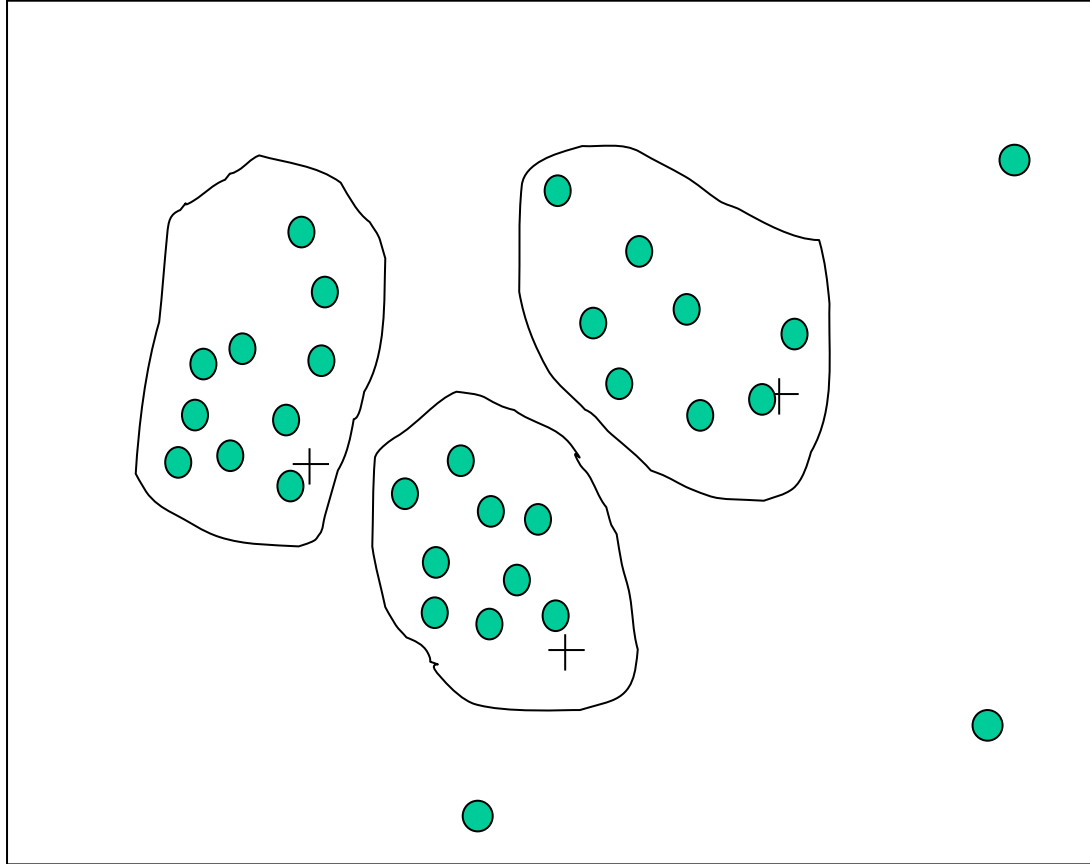
- Sorted data for price: 4, 8, 9, 15, 21, 21, 24, 25, 46, 48, 50, 52
- Partition into (equi-depth) bins:
 - Bin 1: 4, 8, 9, 15
 - Bin 2: 21, 21, 24, 25
 - Bin 3: 46, 48, 50, 52
- Smoothing by bin means:
 - Bin 1: 9, 9, 9, 9
 - Bin 2: 23, 23, 23, 23
 - Bin 3: 49, 49, 49, 49
- Smoothing by bin boundaries:
 - Bin 1: 4, 4, 4, 15
 - Bin 2: 21, 21, 25, 25
 - Bin 3: 46, 46, 52, 52

Smoothing via regression

- For example, feature f has *negative linear correlation* with outdoor temperature
 - $f = a \times \text{outdoor temperature} + b$
 - $\text{Residuals} = \text{measured}_{\text{value}} - \text{predicted}_{\text{value}}$
- Reduce the effect of outdoor temperature on the feature
- De-trend the seasonality of the data

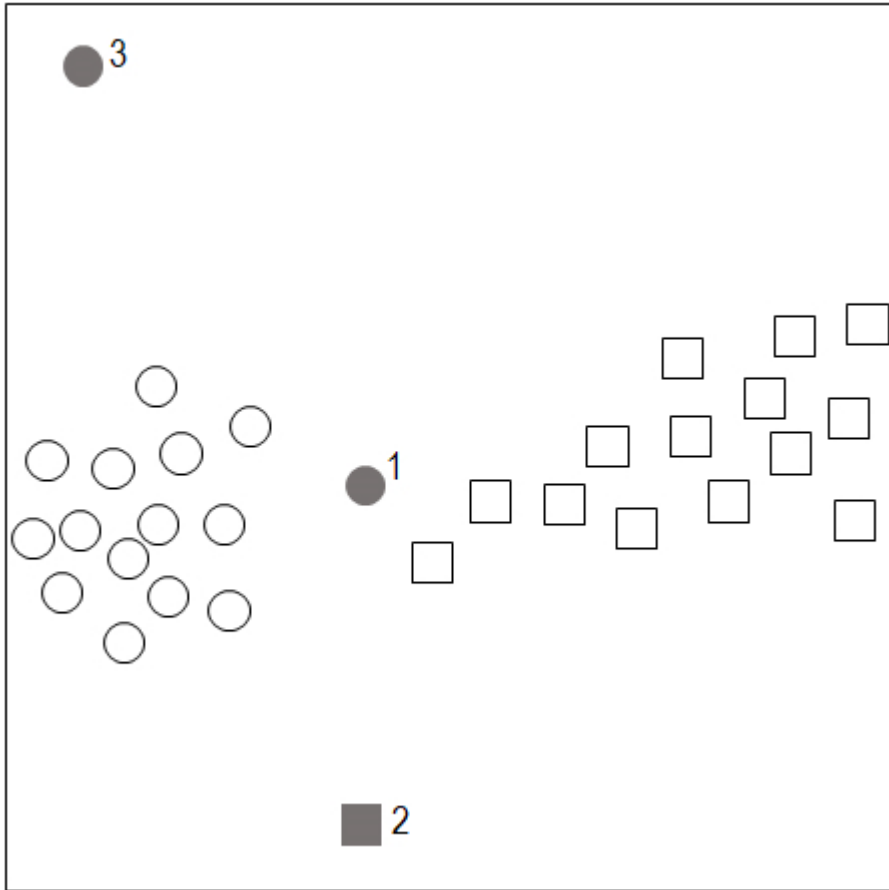


Outlier filtering via clustering analysis



- Cluster the available data objects into groups of similar objects using some suitable distance metric
- Interpret and filter out all distant singleton clusters as outliers

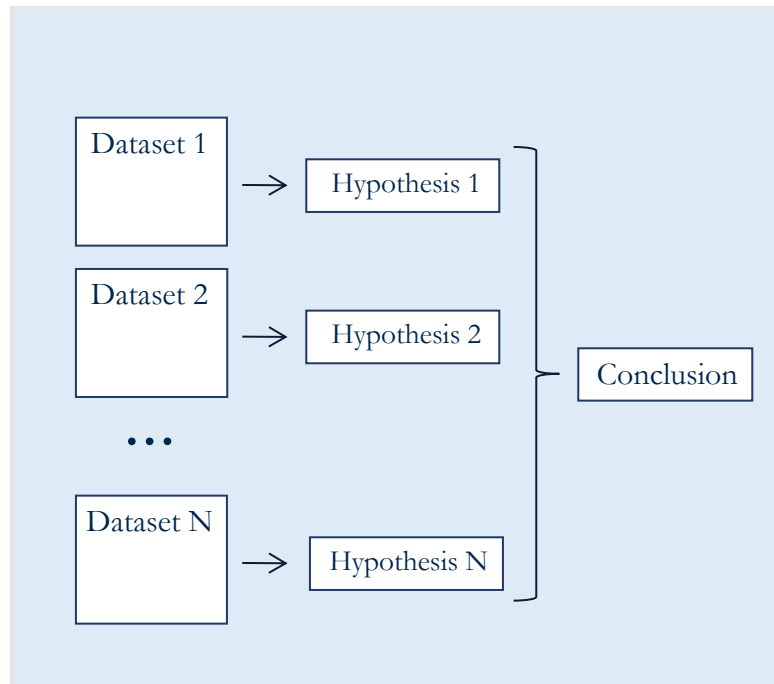
Label noise cleansing



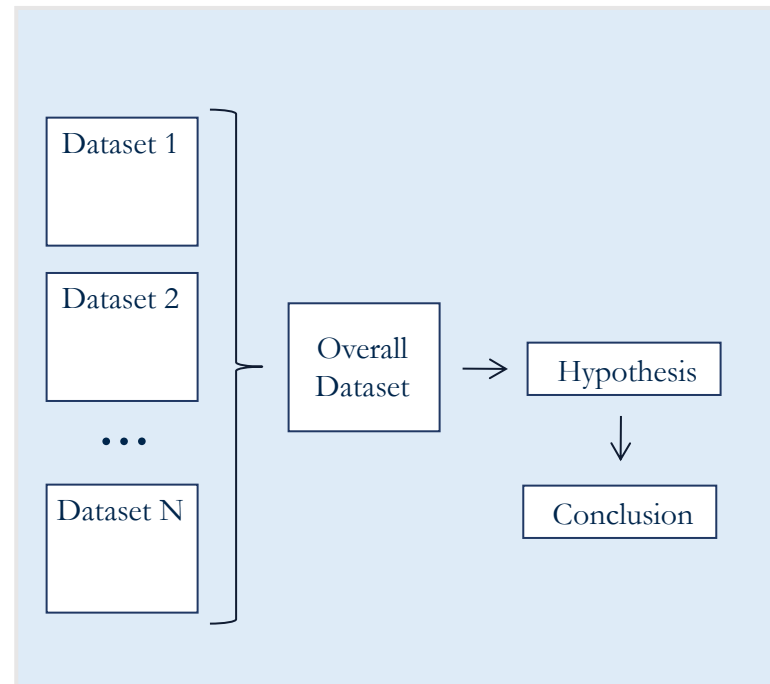
- **Label noise cleansing** methods find suspicious instances taking into account the class label
- **Class outliers** are usually removed prior to training, because they effect the performance of machine learning algorithms

Data integration (1)

Meta-analysis Methods



Inter-study Integration



Data integration is a combination of multiple data sources

Data Integration (2)

- **Inter-study integration methods** combine data from multiple sources into a coherent store
- **Meta-analysis methods**
 - **Schema integration** combines metadata from different sources
 - **Entity identification** identifies real world entities from multiple data sources, e.g., Bill Clinton = William Clinton = B. Clinton
- Detecting and resolving data value conflicts
 - For the same real world entity, attribute values from different sources are different
 - Possible reasons: different representations, different scales, e.g., metric vs. British units

Data Integration (3)

- Redundant data occurs often when integrate multiple databases
 - The same attribute may have different names in different databases
 - One attribute may be a “derived” attribute in another table, e.g., annual revenue
- Redundant data may be able to be detected by correlational analysis
- Careful integration of the data from multiple sources may help reduce/avoid redundancies and inconsistencies and improve mining speed and quality

Data Transformation

- Smoothing: remove noise from data
- Aggregation: summarization
- Generalization: concept hierarchy climbing
- Normalization: scaled to fall within a small, specified range
 - min-max normalization
 - z-score normalization
- Attribute/feature construction
 - New attributes constructed from the given ones

Data Normalization

- **Normalization** is the process of converting an actual range of values, which a numerical feature can take, into a standard range of values, typically in the interval $[-1, 1]$ or $[0, 1]$

- min-max normalization

$$v' = \frac{v - \min_A}{\max_A - \min_A}$$

- Why do we normalize?
 - it can lead to an increased speed of learning

Data Standardization

- **Standardization** is the procedure during which the feature values are rescaled so that they have the properties of a *standard normal distribution* with the mean of 0 and the standard deviation of 1

- z-score normalization

$$v' = \frac{v - \text{mean}_A}{\text{stand_dev}_A}$$

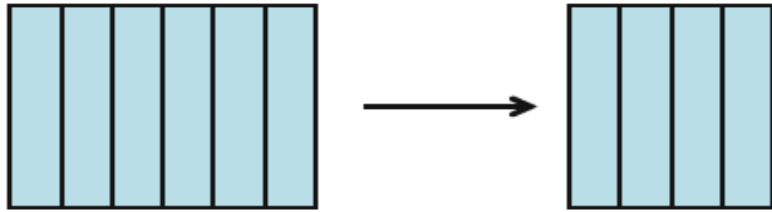
- When we should use normalization and when standardization?
 - No definitive answer
 - Normalize data that is sensitive to outliers
 - Standardization creates a new data not bounded

Data Reduction

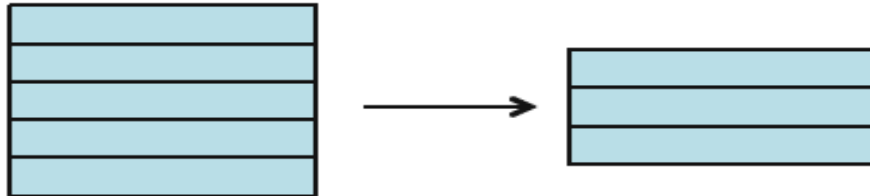
- Complex data analysis/mining may take a very long time to run on the complete data set
- Data reduction
 - Obtains a reduced representation of the data set that is much smaller in volume but yet produces the same (or almost the same) analytical results
- Data reduction strategies
 - Data cube aggregation
 - Dimensionality reduction (feature selection, data compression)
 - Numerosity reduction (histograms, clustering, sampling, parametric methods)
 - Discretization and concept hierarchy generation

Forms of data reduction

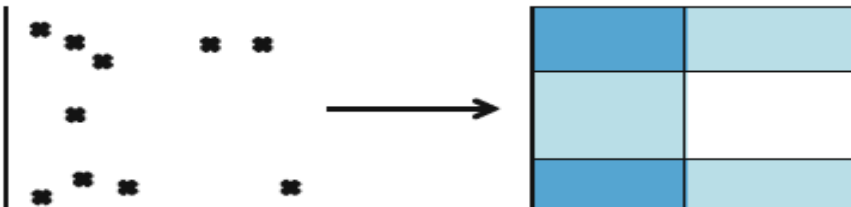
Feature Selection



Instance Selection



Discretization



- Feature selection
 - reduction of the data set by removing irrelevant or redundant features (or dimensions)
- Instance selection/Sampling
 - selection of a subset of the total available data to achieve the original purpose of the data mining application as if the whole data had been used
- Discretization
 - transformation of quantitative data into qualitative data

Feature Selection (1)

- Feature selection:
 - Select a minimum set of features such that the probability distribution of different classes given the values for those features is as close as possible to the original distribution given the values of all features
 - It is easy to understand the reduce number of patterns
- Heuristic methods:
 - step-wise forward selection
 - step-wise backward elimination
 - combining forward selection and backward elimination
 - decision-tree induction

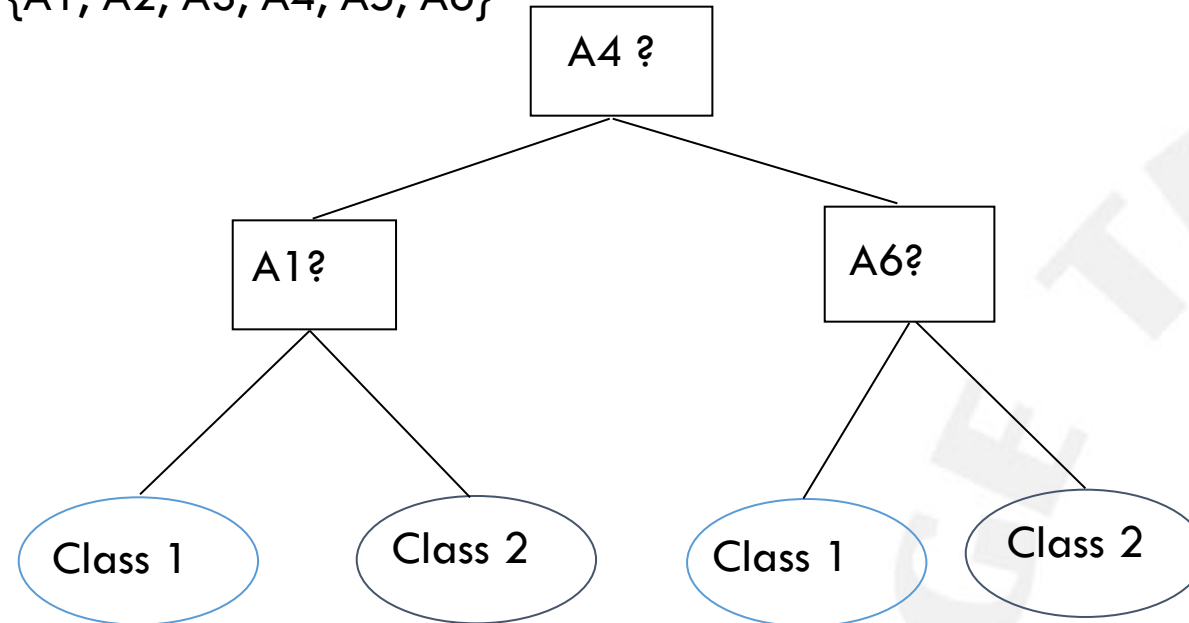
Feature Selection (2)

- Step-wise forward selection:
 - Initial attribute set: $\{X1, X2, X3, X4, X5, X6\}$
 - Original reduce attribute set: $\{ \}$
 - Step-1: $\{X1\}$
 - Step-2: $\{X1, X2\}$
 - Step-3: $\{X1, X2, X5\}$
- Step-wise backward elimination:
 - Initial attribute set: $\{X1, X2, X3, X4, X5, X6\}$
 - Original reduce attribute set: $\{X1, X2, X3, X4, X5, X6\}$
 - Step-1: $\{X1, X2, X3, X4, X5\}$
 - Step-2: $\{X1, X2, X3, X5\}$
 - Step-3: $\{X1, X2, X5\}$

Feature Selection (3)

Example of Decision Tree Induction

Initial attribute set:
 $\{A1, A2, A3, A4, A5, A6\}$



-----> Reduced attribute set: $\{A1, A4, A6\}$

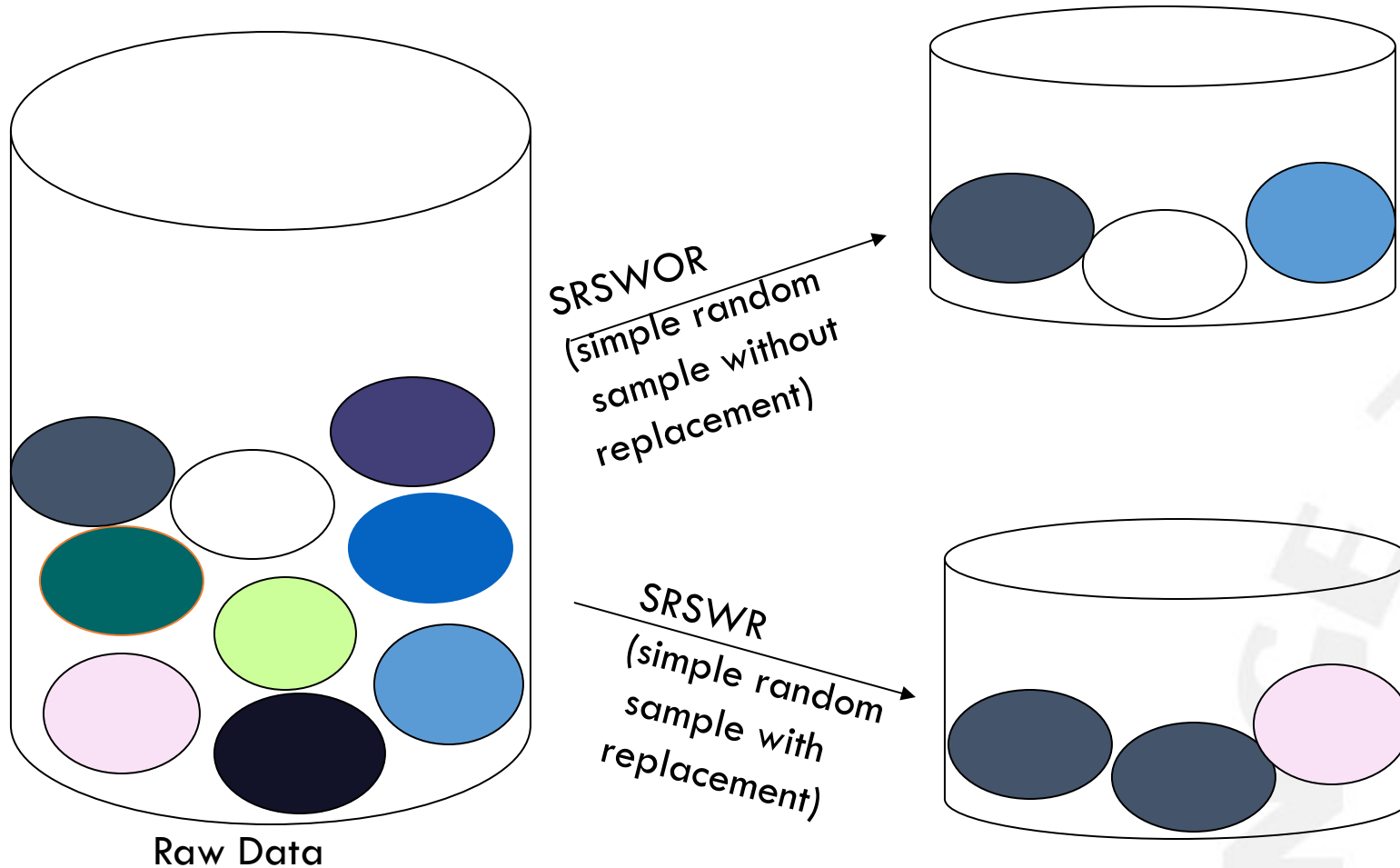
Data Sampling

- **Sampling:** obtaining a small sample s to represent the whole data set S
- Allow a mining algorithm to run in complexity that is potentially sub-linear to the size of the data
- Choose a **representative** subset of the data
 - Simple random sampling may have very poor performance in the presence of skew
- Develop adaptive sampling methods
 - Stratified sampling

Sampling Techniques

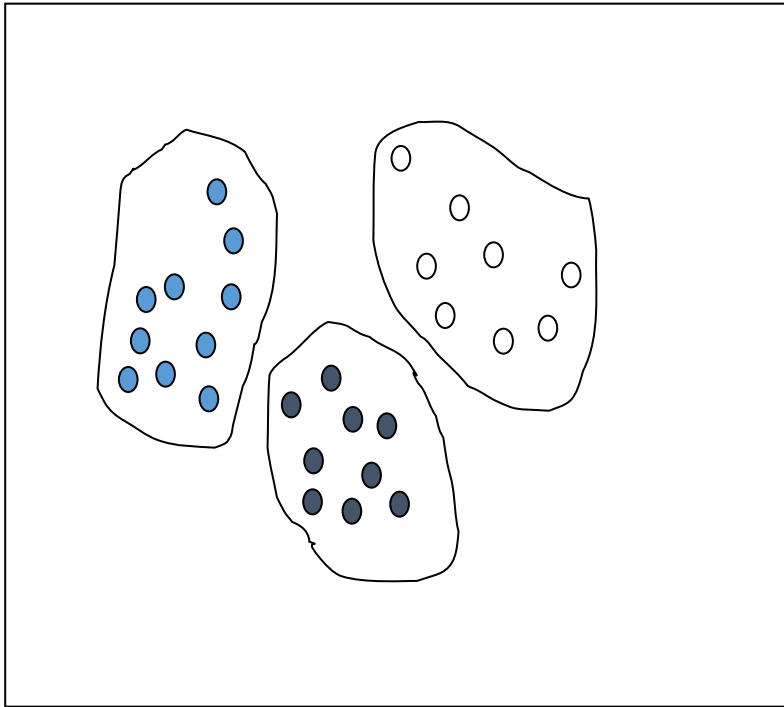
- Simple Random Sample **without Replacement** (SRSWOR)
- Simple Random Sample **with Replacement** (SRSWR)
- Cluster Sample
- Stratified Sample

Simple Random Sampling

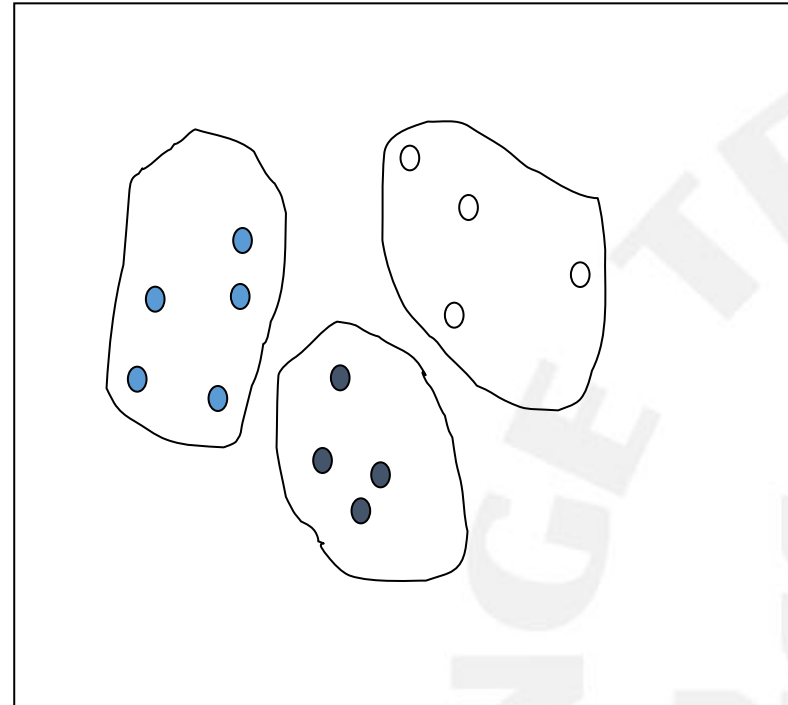


Cluster and Stratified Sampling

Raw Data



Cluster/Stratified Sample



Data reduction via clustering analysis

- Partition data set into clusters, and store only cluster representations (centroids, medoids, exemplars)
- This method can be very effective if data is clustered but not if data is “smeared”
- Data can have hierarchical clustering solution and be stored in multi-dimensional index tree structures

Discretization (1)

- Three types of attributes:
 - Nominal — values from an unordered set
 - Ordinal — values from an ordered set
 - Continuous — real numbers
- Discretization is a process of dividing the range of a continuous attribute into intervals in order to transform numerical attributes into nominal attributes
 - Some classification algorithms only accept categorical (nominal) attributes
 - Reduce data size by discretization
 - Prepare for further analysis

Discretization (2)

- Discretization
 - reduce the number of values for a given continuous attribute by dividing the range of the attribute into intervals
 - Interval labels can then be used to replace actual data values
- Data abstraction/concept hierarchies
 - reduce the data by collecting and replacing low level concepts (such as numeric values for the attribute age) by higher level concepts (such as young, middle-aged, or senior)

Discretization strategies

- Binning
- Histogram analysis
- Clustering analysis
- Entropy-based discretization
- Segmentation by natural partitioning

Simple Discretization Methods: Binning

- Equal-width (distance) partitioning:
 - It divides the range into N intervals of equal size
 - If A and B are the lowest and highest values of the attribute, the width of intervals will be: $W = (B-A)/N$
 - The most straightforward
 - Outliers may dominate presentation
 - Skewed data is not handled well
- Equal-depth (frequency) partitioning:
 - It divides the range into N intervals, each containing approximately same number of samples
 - Good data scaling

Data discretization via histogram analysis

- Histogram is the data representation in terms of frequency
- Histogram analysis is a popular data reduction and discretization technique
- Divide data into buckets and store average (sum) for each bucket
- Related to quantization problems

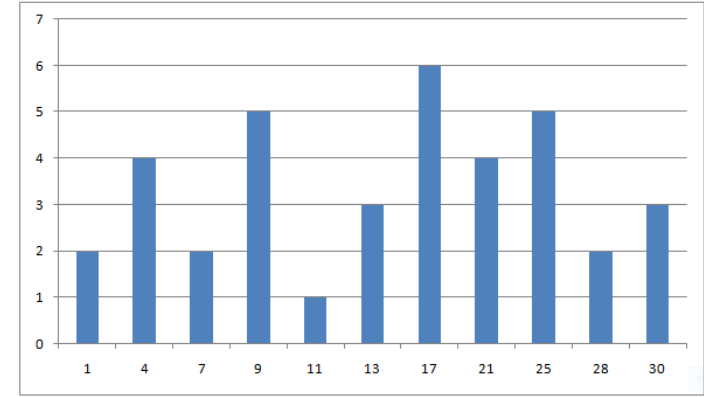


Figure 1 Histogram using price where one bucket represents one value

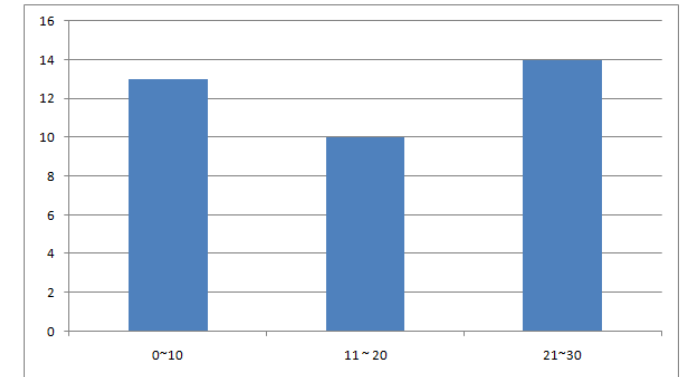


Figure 2: Equal width Histogram

Item price in sorted order:

1, 1, 4, 4, 4, 4, 7, 7, 9, 9, 9, 9, 9, 11, 13, 13, 13, 17, 17, 17, 17, 17, 17, 21, 21, 21, 21, 25, 25, 25, 25, 25, 28, 28, 30, 30, 30.

Entropy-Based Discretization (1)

- Calculate Entropy¹ for the available data.
- For each potential split in data ...
 - Calculate Entropy in each potential bin
 - Find the net entropy for the obtained split
 - Calculate entropy gain
- Select the split with the highest entropy gain
- Recursively (or iteratively in some cases) perform the partition on each split until a termination criteria is met
 - Terminate once you reach a specified number of bins
 - Terminate once entropy gain falls below a certain threshold.

¹Entropy is the measure of disinformation

Entropy-Based Discretization (2)

- Given a set of samples S , if S is partitioned into two intervals S_1 and S_2 using boundary T , the entropy after partitioning is

$$E(S, T) = \frac{|S_1|}{|S|} Ent(S_1) + \frac{|S_2|}{|S|} Ent(S_2)$$

- The boundary that minimizes the entropy function over all possible boundaries is selected as a binary discretization
- The process is recursively applied to partitions obtained until some stopping criterion is met, e.g.,

$$Ent(S) - E(T, S) > \delta$$

- Experiments show that it may reduce data size and improve classification accuracy